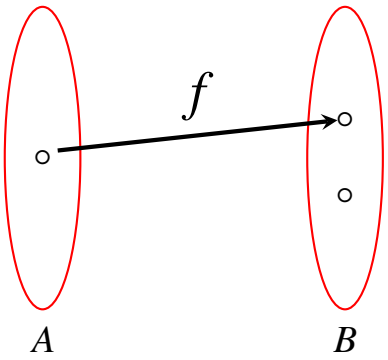
Chapter 3

Linear Transformations

Recall that f is a function from a set A into a set B if $\forall x_1, x_2 \in A$, if $x_1 = x_2$, then $f(x_1) = f(x_2)$.



- The element $f(x)$ of B is called the *image* of x under f.
- $f : A \to B$ $\quad$ f maps A into B.

We are not interested in arbitrary functions from one set into another, but in a special class of functions from one vector space into another vector space. The functions in this class are those that are compatible with the arithmetic operations in each vector space. This is precisely defined in the following theorem.

Definition 3.0.1

A function $T : V \to W$ from a vector space V into a vector space W is called a *linear transformation* if

1. $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$ for all $\vec{u}, \vec{v} \in V$.
2. $T(c\vec{u}) = cT(\vec{u})$ for all $c \in \mathbb{R}$, $\vec{u} \in V$.

Note that the plus sign on the left hand side of the first equation denotes addition in the vector space V, while the plus sign on the right side denotes addition in the vector space W.

Example 3.0.1

Consider $T : \mathbb{R}^2 \to \mathbb{R}^3$ defined by $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 2x \\ x+y \end{bmatrix}$. Show that T is a linear transformation.

1) Let $\begin{bmatrix} x_1 \\ y_1 \end{bmatrix}, \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} \in \mathbb{R}^2$.

$$T\left(\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}\right) = T\left(\begin{bmatrix} x_1 + x_2 \\ y_1 + y_2 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 2(x_1 + x_2) \\ (x_1 + x_2) + (y_1 + y_2) \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 2x_1 \\ x_1 + y_1 \end{bmatrix} + \begin{bmatrix} 0 \\ 2x_2 \\ x_2 + y_2 \end{bmatrix} = T\left(\begin{bmatrix} x_1 \\ y_1 \end{bmatrix}\right) + \left(\begin{bmatrix} x_2 \\ y_2 \end{bmatrix}\right).$$

2) Let $c \in \mathbb{R}$, $\begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$.

$$T\left(c\begin{bmatrix} x \\ y \end{bmatrix}\right) = T\left(\begin{bmatrix} cx \\ cy \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 2cx \\ cx + cy \end{bmatrix} = c\begin{bmatrix} 0 \\ 2x \\ x + y \end{bmatrix} = cT\left(\begin{bmatrix} x \\ y \end{bmatrix}\right).$$

Example 3.0.2

$T : \mathbb{R}^2 \to \mathbb{R}$ defined by $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = xy$ is *not* a linear transformation. Why?

Reason 1: $T\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = T\left(\begin{bmatrix} 3 \\ 4 \end{bmatrix}\right) = 12 \neq 7 = 6 + 1 = T\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) + \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right).$

Reason 2: $T\left(2\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = T\left(\begin{bmatrix} 2 \\ 2 \end{bmatrix}\right) = 4 \neq 2 = 2 \cdot 1 = 2T\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right).$

The following theorem contains the basic properties of linear transformations.

Theorem 3.0.1

Let $T : V \to W$ be a linear transformation of a vector space V into a vector space W. Then

1. $T\left(\vec{0}_V\right) = \vec{0}_W$.
2. $T(-\vec{u}) = -T(\vec{u}), \vec{u} \in V$.
3. $T(c_1\vec{u} + c_2\vec{v}) = c_1T(\vec{u}) + c_2T(\vec{v})$ for all $c_1, c_2 \in \mathbb{R}, \vec{u}, \vec{v} \in V$.

As illustrated by the following examples, if the effect of a linear transformation $T : V \to W$ is known on the elements of a basis for V, then the effect of T on any element of V can be determined.

Example 3.0.3

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation satisfying $T(\vec{u}) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $T(\vec{v}) = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$. Find $T(2\vec{u})$ and $T(5\vec{u} - 3\vec{v})$.

Solution:

$$T(2\vec{u}) = 2T(\vec{u}) = 2\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

$$T(5\vec{u} - 3\vec{v}) = 5T(\vec{u}) - 3\ T(\vec{v}) = 5\begin{bmatrix} 1 \\ 2 \end{bmatrix} - 3\begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix} - 3\begin{bmatrix} 6 \\ 15 \end{bmatrix} = \begin{bmatrix} -1 \\ -5 \end{bmatrix}.$$

Example 3.0.4

Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be a linear transformation satisfying $T\left(\begin{bmatrix}4\\1\end{bmatrix}\right) = \begin{bmatrix}2\\1\\-1\end{bmatrix}$ and $T\left(\begin{bmatrix}3\\1\end{bmatrix}\right) = \begin{bmatrix}4\\-2\\1\end{bmatrix}$. Find $T\left(\begin{bmatrix}x\\y\end{bmatrix}\right)$ for any $\begin{bmatrix}x\\y\end{bmatrix} \in \mathbb{R}^2$.

Solution:

$$\begin{bmatrix}x\\y\end{bmatrix} = c_1\begin{bmatrix}4\\1\end{bmatrix} + c_2\begin{bmatrix}3\\1\end{bmatrix}$$

$$\left[\begin{array}{cc|c}4 & 3 & x\\1 & 1 & y\end{array}\right] \sim \left[\begin{array}{cc|c}1 & 1 & y\\4 & 3 & x\end{array}\right] \begin{matrix}\\R_{12}\end{matrix} \sim \left[\begin{array}{cc|c}1 & 1 & y\\0 & -1 & x-4y\end{array}\right] \begin{matrix}\\R_2 - 4R_1\end{matrix}$$

$$\sim \left[\begin{array}{cc|c}1 & 0 & x-3y\\0 & 1 & 4y-x\end{array}\right] \begin{matrix}R_2 + R_1\\-R_2\end{matrix}$$

Thus $\begin{bmatrix}x\\y\end{bmatrix} = (x-3y)\begin{bmatrix}4\\1\end{bmatrix} + (-x+4y)\begin{bmatrix}3\\1\end{bmatrix}$.

$$T\left(\begin{bmatrix}x\\y\end{bmatrix}\right) = (x-3y)T\left(\begin{bmatrix}4\\1\end{bmatrix}\right) + (-x+4y)T\left(\begin{bmatrix}3\\1\end{bmatrix}\right) = (x-3y)\begin{bmatrix}2\\1\\-1\end{bmatrix} + (-x+4y)\begin{bmatrix}4\\-2\\1\end{bmatrix}$$

$$= \begin{bmatrix}-2x+10y\\3x-11y\\-2x+7y\end{bmatrix} = \begin{bmatrix}-2 & 10\\3 & -11\\-2 & 7\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix}.$$

In the previous examples it can be seen that the given linear transformation can represented by matrix multiplication. In the next section we shall see that every linear transformation from a finite dimensional vector space V into a vector space W can be represented by matrix multiplication.

3.1 Matrix Representation of Linear Transformation

In this section we shall learn how to obtain the matrix representation. We begin by considering the case in which the linear transformation is from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Given a linear transformation $T : V \to W$. Find a matrix A such that

$$T(\vec{x}) = A\vec{x}.$$

Theorem 3.1.1

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. If

$$A = \begin{bmatrix} | & | & & | \\ T\left(\begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}\right) & T\left(\begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}\right) & \cdots & T\left(\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}\right) \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ T(\vec{e}_1) & T(\vec{e}_2) & \cdots & T(\vec{e}_n) \\ | & | & & | \end{bmatrix},$$

then $T(\vec{x}) = A\vec{x}$ for all $\vec{x} \in \mathbb{R}^n$.

Definition 3.1.1

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. The $m \times n$ matrix A such that $T(\vec{x}) = A\vec{x}$ for all $\vec{x} \in \mathbb{R}^n$ is called the *standard matrix* for T.

Example 3.1.1

Consider $T : \mathbb{R}^3 \times \mathbb{R}^4$ defined by $T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_2 \\ x_2 + 2x_3 \\ 0 \\ x_2 \end{bmatrix}$. Find the standard matrix for T

Solution:

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_2 \\ x_2 + 2x_3 \\ 0 \\ x_2 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

Therefore, the standard matrix for T is $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$.

In the previous theorem we found that any linear transformation from $\mathbb{R}^n$ into $\mathbb{R}^m$ can be represented by matrix multiplication. We will show that the same is applied

for any linear transformation from one finite dimensional vector space into another vector space. To construct the matrix representing such a linear transformation we use the concept of coordinate vectors that is already introduced.

Theorem 3.1.2

Let $T : V \to W$ be a linear transformation from a finite dimensional vector space V into a finite dimensional vector space W. Let $\mathcal{B} = \{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ be an ordered basis of V, $\mathcal{B}' = \{\vec{w}_1, \vec{w}_2, \ldots, \vec{w}_m\}$ be an ordered basis of W, and

$$A = \begin{bmatrix} | & | & & | \\ [T(\vec{v}_1)]_{\mathcal{B}'} & [T(\vec{v}_2)]_{\mathcal{B}'} & \cdots & [T(\vec{v}_n)]_{\mathcal{B}'} \\ | & | & & | \end{bmatrix}.$$

Then $[T(\vec{v})]_{\mathcal{B}'} = A\,[\vec{v}]_{\mathcal{B}}$ for every $\vec{v} \in V$.

The basic idea in the above theorem is simple. The matrix A is obtained by computing each of its columns. To get the ith column of A we take the image $T(\vec{v}_i)$ of the ith basis element $\vec{v}_i$ in basis $\mathcal{B}$ of V and compute its coordinate vector $[T(\vec{v}_i)]_{\mathcal{B}'}$ with respect to the basis $\mathcal{B}'$ of W.

Definition 3.1.2

Let $T : V \to W$ be a linear transformation of finite dimensional vector space V into a finite dimensional vector space W. Let $\mathcal{B}$ be an ordered basis of V, $\mathcal{B}'$ be an ordered basis of W. Then matrix ${}_{\mathcal{B}'}A_{\mathcal{B}}$ such that $[T(\vec{v})]_{\mathcal{B}'} = {}_{\mathcal{B}'}A_{\mathcal{B}}\,[\vec{v}]_{\mathcal{B}}$ is called the *matrix representation* of T relative to $\mathcal{B}$ and $\mathcal{B}'$.

Remark: Standard matrix is matrix representation.

Example 3.1.2

Consider a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^2$ defined by $T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} 3x_1 + x_3 \\ x_2 - 2x_3 \end{bmatrix}$. Find the matrix representation of T relative to the ordered bases $\mathcal{B} = \left\{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}\right\}$, $\mathcal{B}' = \left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}\right\}$ of $\mathbb{R}^3$ and $\mathbb{R}^2$, respectively.

Solution:

$$T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 3 \\ 0 \end{bmatrix}, \quad T\left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \quad T\left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$$

Next $\begin{bmatrix} 3 \\ 0 \end{bmatrix} = a_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + a_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 3 \\ 1 \end{bmatrix} = b_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 4 \\ -1 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

$a_1 = -3, \quad a_2 = 3, \quad b_1 = -1, \quad b_2 = 2, \quad c_1 = -6, \quad c_2 = 5.$

Therefore, $\left[T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right)\right]_{\mathcal{B}'} = \begin{bmatrix} -3 \\ 3 \end{bmatrix}$, $\left[T\left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}\right)\right]_{\mathcal{B}'} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$, $\left[T\left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}\right)\right]_{\mathcal{B}'} = \begin{bmatrix} -6 \\ 5 \end{bmatrix}$.

Thus, ${}_{\mathcal{B}'}A_{\mathcal{B}} = \begin{bmatrix} -3 & -1 & -6 \\ 3 & 2 & 5 \end{bmatrix}$. Therefore, $[T(\vec{v})]_{\mathcal{B}'} = {}_{\mathcal{B}'}A_{\mathcal{B}}\,[\vec{v}]_{\mathcal{B}}$.

Now let's additionally consider $\vec{v} = \begin{bmatrix} 4 \\ 2 \\ 3 \end{bmatrix} \in \mathbb{R}^3$. Find $[T(\vec{v})]_{\mathcal{B}'}$. Observe that

$$\begin{bmatrix} 4 \\ 2 \\ 3 \end{bmatrix} = 2\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + 3\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Therefore, $\begin{bmatrix} 4 \\ 2 \\ 3 \end{bmatrix}_{\mathcal{B}} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$. So, $[T(\vec{v})]_{\mathcal{B}'} = \begin{bmatrix} -3 & -1 & -6 \\ 3 & 2 & 5 \end{bmatrix}\begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} -23 \\ 19 \end{bmatrix}$.

3.2 Kernal, Range and Rank-Nullity

Consider the linear transformation $S : \mathbb{R}^n \to \mathbb{R}^m$ defined by

$$S(\vec{x}) = A\vec{x}$$

where A is an $m \times n$ matrix. We found that $rank(A) + nullity(A) = n$. This result has a generalization to linear transformations of a finite dimensional vector space into another vector space.

Let $T : V \to W$ be a linear transformation from a vector space V into a vector space W.

Definition 3.2.1

The *kernel* of T is defined by

$$\text{Ker}T = \left\{ \vec{v} \in V \;\middle|\; T(\vec{v}) = \vec{0}_W \right\}.$$

The *range* of T is defined by

$$\text{range}T = \left\{ \vec{w} \in W \;\middle|\; \vec{w} = T(\vec{v}) \text{ for some } \vec{v} \in V \right\}.$$

Theorem 3.2.1

Let $T : V \to W$ be a linear transformation from a vector space V into a vector space W.

1. ker$T$ is a subspace of V
2. range$T$ is a subspace of W.

Definition 3.2.2

1. The dimension of ker$T$ is called the *nullity of* T, denoted by nullity$T$.
2. The dimension of range$T$ is called the *rank of* T, denoted by rank$T$.

Theorem 3.2.2

If $T : V \to W$ be a linear forms and V is finite dimensional, then rank$T$+nullity$T$=dim$V$.

Recall:

1. T is “onto" if and only if range$T = W$.

2. T is “one-to-one" if and only if $\forall \vec{x}, \vec{y} \in V$ if $T(\vec{x}) = T(\vec{y})$, then $\vec{x} = \vec{y}$.

With these types of linear transformations, we can use this dimension theory to verify the transformations being onto or one-to-one as follows.

Theorem 3.2.3

Let $T : V \to W$ be a linear transformation W is finite dimensional. Then T is “onto" if and only if rank$T$ =dim$W$.

Theorem 3.2.4

Let $T : V \to W$ be a linear transformation W is finite dimensional. Then T is “one-to-one" if and only if ker$T = \left\{\vec{0}_V\right\}$.

Example 3.2.1

Let $\mathbb{R}_2[x]$ be a vector space of all polynomials of degree two over $\mathbb{R}$. Let $T : \mathbb{R}_2[x] \longrightarrow \mathbb{R}^3$ be a libear transformation defined by $T\left(a+bx+cx^2\right) = \begin{bmatrix} a+b \\ b+c \\ 0 \end{bmatrix}$.

Find a basis for ker$T$, a basis for range$T$, and nullity$T$ and rank$T$ and dim$\mathbb{R}_2[x]$.

Solution:

$$\ker T = \left\{ P(x) \in \mathbb{R}_2[x] \;\middle|\; T\left(P(x)\right) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\}.$$

$$T\left(P(x)\right) = T\left(a+bx+cx^2\right) = \begin{bmatrix} a+b \\ b+c \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \implies \begin{array}{l} a+b=0 \implies a=-b=c \\ b+c=0 \implies b=-c,\ c \in \mathbb{R}. \end{array}$$

Therefore, $P(x) = c - cx + cx^2 = c(1-x+x^2)$. Thus, ker$T$ =Span$\left\{1-x+x^2\right\}$. So, nullity$T = 1$.

$$\text{range}T = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \;\middle|\; T\left(P(x)\right) = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ for some } P(x) \in \mathbb{R}_2[x] \right\}$$

Therefore $T\left(a+bx+cx^2\right) = \begin{bmatrix} a+b \\ b+c \\ 0 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} b \\ b \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ c \\ 0 \end{bmatrix} = a \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$.

Therefore range$T$ =Span$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$

Observe that $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$. Therefore, range$T$ =Span$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$

Moreover, $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$ is a basis for range$T$. Therefore, rank$T = 2$.

dim$\mathbb{R}_2[x]$ =rank$T$+nullity$T = 2 + 1 = 3$.

3.3 Eigenvalues and eigenvectors

In this chapter we will be concerned with the equation $A\vec{x} = \lambda\vec{x}$. This equation occurs in many applications of linear algebra such as linear differential equations. If the equation has a nonzero solution $\vec{x}$, then λ is said to be an *eigenvalue* of A and $\vec{x}$ is said to be an *eigenvector* corresponding to λ. Of course, this defined concept can be generalized for any linear transformation.

Let A be a square matrix of order n.

Definition 3.3.1

A real number λ is called an *eigenvalue* of A if $A\vec{x} = \lambda\vec{x}$ has a non-zero solution. The non-zero solution (vector) is called the *eigenvector* corresponding to λ.

Definition 3.3.2

Let $T : V \to W$ be a linear transformation of vector space V into a vector space W having V as a subspace. A number λ is called eigenvalue of T if there exists a non-zero element $\vec{v}$ of V such that

$$T(\vec{v}) = \lambda\vec{v}.$$

This $\vec{v}$ is called the eigenvector corresponding to λ.

Example 3.3.1

Let $A = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$. (a) Is 1 an eigenvalue of A? (b) Is $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ an eigenvector of A?

Solution: (a) $A\vec{x} = \vec{x} \iff A\vec{x} - I_2\vec{x} = \vec{0} \iff (A - I_2)\vec{x} = \vec{0}$.

$$\det(A - I_2) = \begin{vmatrix} 2-1 & 1-0 \\ 4-0 & 2-1 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 4 & 1 \end{vmatrix} = -3 \neq 0.$$

Therefore $A - I_2$ is invertible. Thus, $\vec{x} = \vec{0}$ is the only solution of $A\vec{x} = \vec{x}$. Hence 1 is not an eigenvalue for A.

(b) $A\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \end{bmatrix} = 4\begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

Therefore $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ is an eigenvector for A with the corresponding value $\lambda = 4$.

How to find eigenvalue?

$$\begin{aligned} A\vec{x} = \lambda\vec{x} &\iff A\vec{x} - \lambda\vec{x} = \vec{0}_n \\ &\iff A\vec{x} - \lambda I_n\vec{x} = \vec{0}_n \\ &\iff (A - \lambda I_n)\vec{x} = \vec{0}_n. \end{aligned}$$

Therefore λ is eigenvalue for A if and only if $(A - \lambda I_n)\vec{x} = \vec{0}_n$ has a non-zero solution if and only if $A - \lambda I_n$ is not invertible if and only if $(A - \lambda I_n) = 0$.

The last equation is called "characteristic equation of A" while $\det(A - \lambda I_n)$ is called "characteristic polynomial of A".

Definition 3.3.3

$A_\lambda = Nul(A - \lambda I_n) = \left\{\vec{x} \in \mathbb{R}^n \mid (A - \lambda I_n)\vec{x} = \vec{0}_n\right\}$. If λ is eigenvalue of $\mathbb{R}^n$ containing $\vec{0}_n$ and all eigenvector corresponding to λ. We call this A_λ "the eigenspace of A corresponding to λ".

Example 3.3.2

Let $A = \begin{bmatrix} 3 & 2 \\ 3 & 8 \end{bmatrix}$. Find all eigenvalues for A and a basis for eigenspace corresponding to each eigenvalue.

Solution:

$$\det(A - \lambda I_2) = 0 \iff \begin{vmatrix} 3-\lambda & 2 \\ 3 & 8-\lambda \end{vmatrix} = (3-\lambda)(8-\lambda) - 6 = \lambda^2 - 11\lambda + 18.$$

Thus, $(\lambda - 2)(\lambda - 9) = 0$. Therefore, the eigenvalues are: $\lambda = 2, 9$.

$$A - 2I_2 = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \implies x_1 + 2x_2 = 0 \implies x_1 = -2x_2.$$

Therefore, $\vec{x} \in A_2 \implies \vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -2x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} -2 \\ 1 \end{bmatrix}$.

Therefore, $A_2 = \text{Nul}(A - 2I_2) = \text{Span}\left\{\begin{bmatrix} -2 \\ 1 \end{bmatrix}\right\}$. So, $\left\{\begin{bmatrix} -2 \\ 1 \end{bmatrix}\right\}$ is a basis for A_2.

$$A - 9I_2 = \begin{bmatrix} -6 & 2 \\ 3 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1/3 \\ 0 & 0 \end{bmatrix} \implies x_1 - \frac{1}{3}x_2 = 0 \implies x_1 = \frac{1}{3}x_2.$$

Therefore, $\vec{x} \in A_9 \implies \vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{3}x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} \frac{1}{3} \\ 1 \end{bmatrix}$.

Therefore, $A_9 = \text{Span}\left\{\begin{bmatrix} \frac{1}{3} \\ 1 \end{bmatrix}\right\} = \text{Span}\left\{\begin{bmatrix} 1 \\ 3 \end{bmatrix}\right\}$. So, $\left\{\begin{bmatrix} 1 \\ 3 \end{bmatrix}\right\}$ is a basis for A_9.

3.4 Diagonalization of Matrices

In this section we consider the problem of factoring an $n \times n$ matrix A into a product of the form $P^{-1}DP$, where D is diagonal. This is one of important tools in linear algebra due to its properties. For example, to find an integer power of A will be simplified to find the power of elements in the diagonal entries of D only. We begin with the following theorem.

Theorem 3.4.1

If $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_p$ are eigenvectors corresponding to distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_r$ of matrix A, then $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_r\}$ is linearly independent.

Definition 3.4.1

For square matrix A and B of order n, we say that A *is similar to* B if there exists an invertible matrix P of order n such that $B = P^{-1}AP$.

$$\det B = \det\left(P^{-1}AP\right) = \det(P)^{-1}\det(A)\det(P) = \det A.$$

Remark:

1. If A is similar to B, then $\det B = \det A$.
2. If A is similar to B, then $\det(B - \lambda I_n) = \det\left(P^{-1}AP - \lambda P^{-1}I_nP\right) = \det\left(P^{-1}(A - \lambda I_n)P\right) = \det(A - \lambda I_n)$.

Theorem 3.4.2

If matrix A and B are similar, then A and B have the same characteristic polynomial, hence A and B have the same eigenvalues.

Consider a square matrix A of order n and a diagonal matrix D of order n. Suppose A and D are similar, that is, there exists invertible matrix P, $A = P^{-1}DP$.

$$A^k = \left(P^{-1}DP\right)\left(P^{-1}DP\right)\cdots\left(P^{-1}DP\right) = P^{-1}D^kP.$$

For $D = \begin{bmatrix} d_{11} & & & 0 \\ & d_{22} & & \\ & & \ddots & \\ 0 & & & d_{nn} \end{bmatrix}$, $D^k = \begin{bmatrix} d_{11}^k & & & 0 \\ & d_{22}^k & & \\ & & \ddots & \\ 0 & & & d_{nn}^k \end{bmatrix}$.

Definition 3.4.2

A square matrix A is said to be *diagonalizable* if A is similar to a diagonal matrix.

Suppose that A is similar to $D = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix}$.

Therefore, there exists invertible $P = \begin{bmatrix} \vec{v}_1 & \vec{v}_2 & \dots & \vec{v}_n \end{bmatrix}$ such that $D = P^{-1}AP \iff AP = PD$.

$$A \begin{bmatrix} \vec{v}_1 & \vec{v}_2 & \dots & \vec{v}_n \end{bmatrix} = \begin{bmatrix} \vec{v}_1 & \vec{v}_2 & \dots & \vec{v}_n \end{bmatrix} \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix}$$

$\begin{bmatrix} A\vec{v}_1 & A\vec{v}_2 & \dots & A\vec{v}_n \end{bmatrix} = \begin{bmatrix} \lambda_1\vec{v}_1 & \lambda_2\vec{v}_2 & \dots & \lambda_n\vec{v}_n \end{bmatrix}$. Therefore, $A\vec{v}_i = \lambda_i\vec{v}_i$, where $\vec{v}_i$ is an eigenvector corresponding to eigenvalue λ_i. Need $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ linear independent.

Theorem 3.4.3: Diagonalization Theorem

A square matrix A is diagonalizable, (that is there exists an invertible matrix P and diagonal matrix D such that $A = PDP^{-1}$) if and only if A has n eigenvectors that are linearly independent. If so, $P = \begin{bmatrix} | & | & & | \\ \vec{v}_1 & \vec{v}_2 & \dots & \vec{v}_n \\ | & | & & | \end{bmatrix}$, where $\vec{v}_i's$ are those n eigenvectors and $D = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix}$, where λ_i is the eigenvalue of A that $\vec{v}_i$ corresponds to.

Example 3.4.1

Diagonalize the matrix $A = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}$ and compute A^k for all positive integer k.

Solution:

$$\det(A - \lambda I_2) = \begin{vmatrix} 7-\lambda & 2 \\ -4 & 1-\lambda \end{vmatrix} = ((7-\lambda)(1-\lambda) + 8 = \lambda^2 - 8\lambda + 15 = 0.$$

Thus $(\lambda - 3)(\lambda - 5) = 0$. Therefore, $\lambda = 3,\ 5$ $(eigenvalues)$.

For $\lambda = 3$, $A - 3I_2 = \begin{bmatrix} 4 & 2 \\ -4 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 1/2 \\ 0 & 0 \end{bmatrix} \implies x_1 + \frac{1}{2}x_2 = 0 \implies x_1 = -\frac{1}{2}x_2.$

Therefore, $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2}x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} -\frac{1}{2} \\ 1 \end{bmatrix}.$

Therefore, $A_3 = \text{Nul}(A - 3I_2) = \text{Span}\left\{\begin{bmatrix} -1/2 \\ 1 \end{bmatrix}\right\} = \text{Span}\left\{\begin{bmatrix} -1 \\ 2 \end{bmatrix}\right\}.$ So, $\vec{v}_1 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$

For $\lambda = 3$, $A - 5I_2 = \begin{bmatrix} 2 & 2 \\ -4 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \implies x_1 + x_2 = 0 \implies x_1 = -x_2.$

Therefore, $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$

Therefore, $A_5 = \text{Nul}(A - 5I_2) = \text{Span}\left\{\begin{bmatrix} -1 \\ 1 \end{bmatrix}\right\}.$ So, $\vec{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

Since $\vec{v}_1$ and $\vec{v}_2$ correspond to distinct eigenvalues, $\{\vec{v}_1, \vec{v}_2\}$ linearly independent.

$$\text{Thus,} \quad D = \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}, \quad P = \begin{bmatrix} -1 & -1 \\ 2 & 1 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} 1 & 1 \\ -2 & -1 \end{bmatrix}$$

$$\text{Therefore,} \quad A = PDP^{-1} = \begin{bmatrix} -1 & -1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -2 & -1 \end{bmatrix}$$

$$A^k = \begin{bmatrix} -1 & -1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3^k & 0 \\ 0 & 5^k \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -2 & -1 \end{bmatrix} = \begin{bmatrix} -3^k + 2 \cdot 5^k & -3^k + 5^k \\ 2 \cdot 3^k - 2 \cdot 5^k & 2 \cdot 3^k - 5^k \end{bmatrix}.$$

Theorem 3.4.4

If A is a square matrix of order n with n *distinct* eigenvalues, then A is diagonalizable.

What happens if A has some repeated eigenvalues? (i.e., the characteristic polynomial of A has some repeated roots)

Theorem 3.4.5

Let A be a aquare matrix of order n having distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_p$, $p < n$.

1. nullity$(A - \lambda_k I_n) \leq$ number of repeated times of λ_k for all $k = 1, \cdots, p$.
2. A is diagonalizable iff nullity$(A - \lambda_k I_n) = n$ iff nullity$(A - \lambda_k I_n) =$ number of repeated times of λ_k for all $k = 1, \cdots, p$.
3. If A is diagonalizable and $\mathcal{B}_k$ is a basis of A_{λ_k} for all $k = 1, 2 \cdots, p$, then $\mathcal{B} = \mathcal{B}_1 \cup \mathcal{B}_2 \cup \mathcal{B}_3 \cup \cdots \cup \mathcal{B}_p$ is a basis for $\mathbb{R}^n$.

Example 3.4.2

Determine if the following matrices are diagonalizable.

1. $A = \begin{bmatrix} 5 & -6 & -6 \\ -1 & & 4 \quad 2 \\ 3 & -6 & -4 \end{bmatrix}$ $\qquad A = \begin{bmatrix} 3 & 1 & -1 \\ 2 & 2 & -1 \\ 2 & 2 & 0 \end{bmatrix}$

Solution:

1. $\det(A - \lambda I_3) = \begin{vmatrix} 5-\lambda & -6 & -6 \\ -1 & 4-\lambda & 2 \\ 3 & -6 & -4-\lambda \end{vmatrix}$

$= (5-\lambda)(4-\lambda)(-4-\lambda) - 36 - 36 + 18(4-\lambda) + 12(5-\lambda) - 6(-4-\lambda)$

$= -\lambda^3 + 5\lambda^2 - 8\lambda + 4 = 0$

Thus $\quad -(\lambda-1)(\lambda-2)^2 = 0.$ Therefore, eigenvalues are $\lambda = 1, 2, 2.$

For $\lambda = 1$, nullity$(A - I_3) = 1$ because there is any one $\lambda = 1$ as eigenvalue.

$\lambda = 1 \implies \det(A - I_3) = 0 \implies \text{Nul}(A - I_3) \neq \left\{\vec{0}\right\}.$

For $\lambda = 1$, nullity$(A - I_3) \leq 2$. $\quad A - 2I_3 = \begin{bmatrix} 3 & -6 & -6 \\ -1 & 2 & 2 \\ 3 & -6 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$

Therefore $x_1 - 2x_2 - 2x_3 = 0 \implies x_1 = 2x_1 + 2x_3.$

$\vec{x} \in \text{Nul}(A - 2I_3) \implies \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2x_2 + 2x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$

Therefore, $A_2 = \text{Nul}(A - sI_3) = \text{Span}\left\{\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}\right\}$. So, nullity$(A - 2I_3) = 2.$

nullity$(A - I_3)$+ nullity$(A - 2I_3) = 1 + 2 = 3 = n.$ Therefore A is diagonalizable.

2. $\det(A - \lambda I_3) = \begin{vmatrix} 3-\lambda & 1 & -1 \\ 2 & 2-\lambda\lambda & -1 \\ 2 & 2 & -\lambda \end{vmatrix} = -\lambda^3 + 5\lambda^2 - 8\lambda + 4 = -(\lambda-1)(\lambda-2)^2 = 0$

Therefore, eigenvalues are $\lambda = 1, 2, 2$

For $\lambda = 1$, $\quad$ nullity$(A - I_3) = 1$

For $\lambda = 2$, $\quad A - 2I_3 = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 0 & -1 \\ 2 & 2 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1/2 \\ 0 & 1 & -1/2 \\ 0 & 0 & 0 \end{bmatrix}.$ Therefore rank$(A - 2I_3) = 2.$

By rank$(A - 2I_3)$+nullity$(A - 2I_3) = 3$, we have 2+nullity$(A - 2I_3) = 3$.

Therefore nullity$(A - 2I_3) = 1 \neq$ number of times of $\lambda = 2, 2$. Therefore, A is not diagonalizable.

Summary for diagonalization of A

1. Find λ from $\det(A - \lambda I_n) = 0$, λ can be repeated.
2. Find basis and dimension of eigenspaces corresponding to each λ.
3. Use Theorems to determine if A is diagonalizable.
4. If so, write $A = PDP^{-1}$, D = diagonal matrix from all eigenvalues P from eigenvectors correspond to each eigenvalues.

Diagonalization of Linear Transformations

Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a linear transformation. T is *diagonalizable* if there exists a basis $\mathcal{B}'$ of $\mathbb{R}^n$ such that the matrix representation ${}_{\mathcal{B}'}A_{\mathcal{B}'}$ of T is diagonalizable.

Theorem 3.4.6

Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a linear transformation. T is diagonalizable if and only if A is diagonalizable, that is, there exists an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$. Moreover, the basis $\mathcal{B}'$ formed by columns of P makes ${}_{\mathcal{B}'}A_{\mathcal{B}'} = D$.

Example 3.4.3

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation defined by $T(x_1, x_2) = (7x_1 + 2x_2, -4x_1 + x_2)$, for all $(x_1, x_2) \in \mathbb{R}^2$. Find a basis $\mathcal{B}'$ for $\mathbb{R}^2$ for which ${}_{\mathcal{B}'}A_{\mathcal{B}'}$ of T is diagonalizable (If exists).

Solution:

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 7x_1 + 2x_2 \\ -4x_1 + x_2 \end{bmatrix} = x_1 \begin{bmatrix} 7 \\ 4 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}. \quad \Longrightarrow \quad A = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}.$$

$$\det(A - \lambda I_2) = \begin{vmatrix} 7 - \lambda & 2 \\ -4 & 1 - \lambda \end{vmatrix} = (7 - \lambda)(1 - \lambda) + 8 = \lambda^2 - 8\lambda + 15$$

Thus, $(\lambda - 3)(\lambda - 5) = 0$. Therefore, eigenvalues $\lambda = 3, 5$. So, A is diagonalizable.

Moreover, bases for eigenspaces corresponding to $\lambda = 3$ and $\lambda = 5$ are $\left\{\begin{bmatrix} -1 \\ 2 \end{bmatrix}\right\}$ and $\left\{\begin{bmatrix} -1 \\ 1 \end{bmatrix}\right\}$, respectively.

From Theorem above, $\mathcal{B}' = \left\{\begin{bmatrix} -1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix}\right\}$ and ${}_{\mathcal{B}'}A_{\mathcal{B}'} = \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}$.

3.5 Jordan Canonical Form

Note that a square matrix A of order n may not be diagonalizable. However, it is still possible to find a matrix M so that $M^{-1}AM$ has an exceptionally simple form.

Definition 3.5.1

An $m \times m$ matrix of the form $\begin{bmatrix} a & 1 & 0 & \cdots & 0 & 0 \\ 0 & a & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & 1 & 0 \\ 0 & 0 & 0 & \cdots & a & 1 \\ 0 & 0 & 0 & \cdots & 0 & a \end{bmatrix}$ is called a *Jordan block* correspond to a.

Example 3.5.1

$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$ is a 3×3 Jordan block correspond to 2.

Definition 3.5.2

An $n \times n$ matrix J is said to be in *Jordan canonical form* if it is a diagonal matrix of the form $\begin{bmatrix} M_1 & 0 & \cdots & 0 \\ 0 & M_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & M_k \end{bmatrix}$, where $M_1, M_2, \ldots, M_k$ are

1) all Jordan blocks, or

2) M_1 is diagonal matrix and $M_2, \ldots, M_k are$ are Jordan blocks.

The next theorem, which extends the notion of a matrix being diagonalizable, is one of the fundamental theorems of matrix theory.

Theorem 3.5.1

Every $n \times n$ matrix is similar to a matrix in Jordan canonical form

Remark: A diagonal matrix is a in Jordan canonical form.

To proceed on find such a form, we need the following definition.

Definition 3.5.3

An n-vector $\vec{x}$ is called a *generalised eigenvector of rank p* correspond to λ if $(A-\lambda I)^p\vec{x} = \vec{0}$ and $(A-\lambda I)^{p-1}\vec{x} \neq \vec{0}$.

Theorem 3.5.2

If an $n \times n$ matrix A has t linearly independent eigenvectors, then A is similar to a matrix J, that is, in Jordan Canonical Form, with t blocks on the diagonal:

$$J = M^{-1}AM = \begin{bmatrix} M_1 & & & \\ & M_2 & & \\ & & \ddots & \\ & & & M_t \end{bmatrix}.$$

Each block has one eigenvector, one eigenvalue (M_i is Jordan block) $M_i = \begin{bmatrix} \lambda_i & 1 & & \\ & \lambda_i & 1 & \\ & & \ddots & 1 \\ & & & \lambda_i \end{bmatrix}$.

The same λ_i will appear in several blocks if it has several independent eigenvectors. Moreover, M consists of n generalized eigenvectors which are linearly independent.

3.6 Quadratic forms

A function of the form

$$F(x, y) = ax^2 + bxy + cy^2$$

is called a *quadratic form* in x and y. Quadratic forms arise in many areas such as economics, geometry, mechanics, and statistics. Before defining the quadratic form in general, we calculate the following example.

Example 3.6.1

Let $\vec{x} = \begin{bmatrix} x \\ y \end{bmatrix}$. Find $\vec{x}^T A\vec{x}$ for

1. $A = \begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix}$.

   Solution: $\vec{x}^T A\vec{x} = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 2x^2 - 3y^2.$

2. $A = \begin{bmatrix} 3 & -2 \\ -2 & 1 \end{bmatrix}$.

   Solution: $\vec{x}^T A\vec{x} = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 2x^2 - 4xy + y^2.$

Definition 3.6.1

Let A be a symmetric matrix. We call a function $Q : \mathbb{R}^n \to \mathbb{R}$ defined by

$$Q(\vec{x}) = \vec{x}^T A\vec{x}$$

for all $\vec{x} \in \mathbb{R}^n$, a *quadratic form*. We call this matrix A a *matrix of the quadratic form* Q.

Example 3.6.2

Given that $Q(x)) = 3x_1^2 - 5x_2^2 + x_3^2 - 2x_1x_2 + x_2x_3$ for all $\vec{x} \in \mathbb{R}^3$. Show that Q is a quadratic form and find the matrix of Q.

Solution:

Let $A = \begin{bmatrix} a & d & e \\ d & b & f \\ e & f & c \end{bmatrix}$ be a symmetric matrix such that $Q(\vec{x}) = \vec{x}^T A\vec{x}$.

Thus $\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} a & d & e \\ d & b & f \\ e & f & c \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = Q(\vec{x})$

$$ax_1^2 + bx_2^2 + cx_3^2 + 2dx_1x_2 + 2ex_1x_3 + 2fx_2x_3 = 3x_1^2 - 5x_2^2 + x_3^2 - 2x_1x_2 + x_2x_3$$

Therefore, $a = 3,\ b = -5,\ c = 1,\ d = -1,\ e = 0,\ f = \frac{1}{2}$. So, $A = \begin{bmatrix} 3 & -1 & 0 \\ -1 & -5 & 1/2 \\ 0 & 1/2 & 1 \end{bmatrix}$.

Remark: 1. Quadratic form on $\mathbb{R}^2$

$$Q(\vec{x}) = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a & c \\ c & b \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = ax_1^2 + bx_2^2 + 2cx_1x_2.$$

2. Quadratic form on $\mathbb{R}^3$

$$Q(\vec{x}) = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} a & d & e \\ d & b & f \\ e & f & c \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = ax_1^2 + bx_2^2 + cx_3^2 + 2dx_1x_2 + 2ex_1x_3 + 2fx_2x_3.$$

We call "cx_ix_j", $i \neq j$ a cross-product term.

Let $\vec{x} \in \mathbb{R}^n$. We call $x = \vec{P\vec{y}}$ or $\vec{y} = P^{-1}\vec{x}$, where P is an invertible matrix, $\vec{y} \in \mathbb{R}^n$, *change of variables*.

Let A be a symmetric matrix. Then find an "*orthonormal matrix*" P and a diagonal matrix D such that $P^TAP = D$. So, we can change variable $\vec{x} = P\vec{y}$ for which

$$\vec{x}^TA\vec{x} = (P\vec{y})^T A (P\vec{y}) = \vec{y}^T \left(P^TAP\right) \vec{y} = \vec{y}^T D\vec{y}$$

while there is no cross-product term in variable $\vec{y}$.
Note that P is orthonormal vectors $P^TP = PP^T = I$. Therefore $P^T = P^{-1}$.

Example 3.6.3

Let $Q(x_1, x_2) = 3x_1^2 + 6x_2^2 - 4x_1x_2$ be a quadratic form. Change variable $\vec{x} = P\vec{y}$ such that Q has no cross-product term.

Solution:

$Q(\vec{x}) = \vec{x}^T A\vec{x}$, where $A = \begin{bmatrix} 3 & -2 \\ -2 & 6 \end{bmatrix}$.

$\det(A - \lambda I_2) = \begin{vmatrix} 3-\lambda & -2 \\ -2 & 6-\lambda \end{vmatrix} = \lambda^2 - 9\lambda + 14 = 0. \quad (\lambda - 2)(\lambda - 7) = 0.$

Therefore, eigenvalues are $\lambda = 2, 7$.

For $\lambda = 2$, $A - 2I_2 = \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix} \implies x_1 - 2x_2 = 0 \implies x_1 = 2x_2.$

Therefore, $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. So, $\text{Nul}(A - 2I_2) = \text{Span}\left\{\begin{bmatrix} 2 \\ 1 \end{bmatrix}\right\}$.

For $\lambda = 7$, $A - 7I_2 = \begin{bmatrix} -4 & -2 \\ -2 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1/2 \\ 0 & 0 \end{bmatrix} \implies x_1 + \frac{1}{2}x_2 = 0 \implies x_2 = -2x_1.$

Therefore, $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ -2x_1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. So, $\text{Nul}(A - 7I_2) = \text{Span}\left\{\begin{bmatrix} 1 \\ -2 \end{bmatrix}\right\}$.

Normalize vectors: $\vec{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$.

Then $\|\vec{u}\| = \sqrt{2^2 + 1^2} = \sqrt{5}$ and $\|\vec{v}\| = \sqrt{1^2 + (-2)^2} = \sqrt{5}$.

Therefore, $P = \begin{bmatrix} 2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & -2/\sqrt{5} \end{bmatrix}$ and $D = \begin{bmatrix} 2 & 0 \\ 0 & 7 \end{bmatrix}$.

Let $\vec{x} = P\vec{y}$. Then $Q(x_1, x_2) = \vec{x}^T A\vec{x} = \cdots = \vec{y}^T D\vec{y}$.

Therefore, $\begin{bmatrix} y_1 & y_2 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 7 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = 2y_1^2 + 7y_2^2 = Q(y_1, y_2)$.

$3x_1^2 + 6x_2^2 - 4x_1x_2 = 14 \implies 2y_1^2 + 7y_2^2 = 14.$

One of the properties that is commonly desired of a quadratic form is described in the following definition.

Definition 3.6.2

We call the quadratic form Q

1. "positive definite (positive semidefinite, resp)" if

$$Q(\vec{x}) > 0 \text{ for all } \vec{x} \neq 0 \quad (Q(\vec{x}) \geq 0 \text{ all } \vec{x}).$$

2. "negative definite (negative semidefinite, resp)" if

$$Q(\vec{x}) < 0 \text{ for all } \vec{x} \neq 0 \quad (Q(\vec{x}) \leq 0 \text{ all } \vec{x}).$$

3. "indefinite" if neither (1) nor (2).

Theorem 3.6.1

Let A be an invertible symmetric matrix. Then the quadratic form $\vec{x}^T A\vec{x}$ is

1. positive definite if and only if all eigenvalues of A are positive.
2. negative definite if and only if all eigenvalues of A are negative.
3. indefinite if and only if A has positive and negative eigenvalues.

3.7 Exercises 3

This exercise will be use in the group discussion for final exam preparation.

1. Determine whether the given function $T : \mathbb{R}^2 \to \mathbb{R}^2$ is a linear transformation. If it is not, show why.

   a) $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2x + y \\ x - 3y \end{bmatrix}$

   b) $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} \cos x \\ \sin y \end{bmatrix}$

2. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \text{ and } T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$$

   a) Find $T\left(\begin{bmatrix} -2 \\ 4 \end{bmatrix}\right)$

   b) Find $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right)$

   c) Find a matrix A such that $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = A \begin{bmatrix} x \\ y \end{bmatrix}$

3. Let $B = \left\{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right\}$ and $B' = \left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix}\right\}$ be ordered bases for $\mathbb{R}^3$ and $\mathbb{R}^2$ respectively. Find the matrix representation of the linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^2$ defined by

$$T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 4 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

4. Given a linear transformation $T : \mathbb{R}^4 \to \mathbb{R}^3$ defined by

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}\right) = \begin{bmatrix} 2x_2 - 5x_4 \\ x_1 + 3x_2 \\ x_3 \end{bmatrix}$$

Find bases for the kernel and range of this linear transformation. Also verify that the conclusion of the Rank-Nullity theorem holds for this T.

5. Find the eigenvalues and associated eigenvectors of the given matrices.

   a) $\begin{bmatrix} 0 & 3 \\ 3 & 0 \end{bmatrix}$

   b) $\begin{bmatrix} 2 & 0 & -2 \\ 3 & 4 & 3 \\ 2 & 0 & 6 \end{bmatrix}$

6. Show that the eigenvalues of a diagonal matrix are the numbers on the diagonal.

7. Show that if λ is an eigenvalue of A and k is a positive integer, then λ^k is an eigenvalue of A^k.

8. Compute the eigenspace for each eigenvalue of $A = \begin{bmatrix} 5 & -1 & 1 \\ 0 & 4 & 0 \\ 1 & -1 & 5 \end{bmatrix}$.
   Also give the multiplicities of each eigenvalue.

9. Determine whether the matrix A is diagonalizable. If it is, find an invertible matrix P such that $P^{-1}AP$ is a diagonal matrix and determine $P^{-1}AP$.

   a) $\begin{bmatrix} 4 & 1 \\ -1 & 2 \end{bmatrix}$

   b) $\begin{bmatrix} 2 & 0 & 1 \\ 0 & 5 & 0 \\ 3 & 0 & 4 \end{bmatrix}$

10. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear transformation defined by

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 3 & 4 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

    Find a basis for $\mathbb{R}^2$ for which the matrix representation for T is diagonal.

1. Determine whether the given function $T : \mathbb{R}^2 \to \mathbb{R}^2$ is a linear transformation. If it is not, show why.

a) $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2x + y \\ x - 3y \end{bmatrix}$

b) $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} \cos x \\ \sin y \end{bmatrix}$

Sol!! a) ให้ $u = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$, $v = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$, $c \in \mathbb{R}$

พิจารณา $T\left(\begin{bmatrix} x_1 \\ y_1 \end{bmatrix}\right) + T\left(\begin{bmatrix} x_2 \\ y_2 \end{bmatrix}\right)$

$= \begin{bmatrix} 2x_1+y_1 \\ x_1-3y_1 \end{bmatrix} + \begin{bmatrix} 2x_2+y_2 \\ x_2-3y_2 \end{bmatrix}$

$= \begin{bmatrix} 2(x_1+x_2)+(y_1+y_2) \\ (x_1-x_2)-3(y_1+y_2) \end{bmatrix}$

$= T\left(\begin{bmatrix} x_1+x_2 \\ y_1+y_2 \end{bmatrix}\right)$ เป็นจริง

พิจารณา $cT\left(\begin{bmatrix} x_1 \\ y_1 \end{bmatrix}\right) = c\begin{bmatrix} 2x_1+y_1 \\ x_1-3y_1 \end{bmatrix}$

$= \begin{bmatrix} 2cx_1+cy_1 \\ cx_1-3cy_1 \end{bmatrix}$

$= T\left(\begin{bmatrix} cx_1 \\ cy_1 \end{bmatrix}\right)$

$\therefore$ ข้อ a) ฟังก์ชัน T มีการแปลงเชิงเส้น # Ans.

b) ให้ $x = y = \frac{\pi}{6}$, $c = 2$

พิจารณา $cT\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = 2\begin{bmatrix} \cos\frac{\pi}{6} \\ \sin\frac{\pi}{6} \end{bmatrix} = \begin{bmatrix} 2\sqrt{3} \\ 1 \end{bmatrix}$

พิจารณา $T\left(\begin{bmatrix} cx \\ cy \end{bmatrix}\right) = \begin{bmatrix} \cos 2(\frac{\pi}{6}) \\ \sin 2(\frac{\pi}{6}) \end{bmatrix} = \begin{bmatrix} 1/2 \\ \sqrt{3}/2 \end{bmatrix}$

จะเห็นว่า $cT\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) \neq T\left(\begin{bmatrix} cx \\ cy \end{bmatrix}\right)$

$\therefore$ ข้อ b) ฟังก์ชัน T ไม่มีการแปลงเชิงเส้น เพราะไม่มีสมบัติปิดการคูณ # Ans.

2. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \text{ and } T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$$

a) Find $T\left(\begin{bmatrix} -2 \\ 4 \end{bmatrix}\right)$

b) Find $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right)$

c) Find a matrix A such that $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = A\begin{bmatrix} x \\ y \end{bmatrix}$

Sol!! จากโจทย์ ให้ $\begin{bmatrix} x \\ y \end{bmatrix} = c_1\begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $\longrightarrow$ ทำให้ได้ $c_1 = x$, $c_2 = y$

จะได้ $\begin{bmatrix} x \\ y \end{bmatrix} = x\begin{bmatrix} 1 \\ 0 \end{bmatrix} + y\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = xT\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) + yT\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = x\begin{bmatrix} 1 \\ 3 \end{bmatrix} + y\begin{bmatrix} -2 \\ 4 \end{bmatrix}$

ทำให้ได้ว่า $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x-2y \\ 3x+4y \end{bmatrix}$

a) ทำให้ได้ $T\left(\begin{bmatrix} -2 \\ 4 \end{bmatrix}\right) = \begin{bmatrix} -2-2(4) \\ 3(-2)+4(4) \end{bmatrix}$

$\therefore$ $T\left(\begin{bmatrix} -2 \\ 4 \end{bmatrix}\right) = \begin{bmatrix} -10 \\ 10 \end{bmatrix}$ # Ans.

b) ทำให้ได้ $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x-2y \\ 3x+4y \end{bmatrix}$ # Ans.

c) ทำให้ได้ $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x-2y \\ 3x+4y \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$

$\therefore$ $A = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$ # Ans.

3. Let $B = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$ and $B' = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}$ be ordered bases for $\mathbb{R}^3$ and $\mathbb{R}^2$ respectively. Find the matrix representation of the linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^2$ defined by

$$T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 4 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

Sol!! จากโจทย์ ที่เรารู้ว่า $T\left(\begin{bmatrix}1\\0\\1\end{bmatrix}\right) = \begin{bmatrix}1\\3\end{bmatrix} = a_1\begin{bmatrix}1\\1\end{bmatrix} + a_2\begin{bmatrix}-1\\1\end{bmatrix}$ (เพราะ $\begin{bmatrix}1(1)+2(0)-1(1)\\3(1)+4(0)-2(1)\end{bmatrix} = \begin{bmatrix}1\\3\end{bmatrix}$) $\to$ ใช้ Row operation :

$$\left[\begin{array}{cc|c}1&-1&0\\1&1&1\end{array}\right] \sim \left[\begin{array}{cc|c}1&-1&0\\0&2&1\end{array}\right]_{R_2-R_1} \sim \left[\begin{array}{cc|c}1&-1&0\\0&1&1/2\end{array}\right]_{\frac{1}{2}R_2} \sim \left[\begin{array}{cc|c}1&0&1/2\\0&1&1/2\end{array}\right]^{R_1-R_2}$$

ทำให้ได้ว่า $a_1 = a_2 = \frac{1}{2}$

ที่เรารู้ว่า $T\left(\begin{bmatrix}0\\1\\0\end{bmatrix}\right) = \begin{bmatrix}2\\4\end{bmatrix} = b_1\begin{bmatrix}1\\1\end{bmatrix} + b_2\begin{bmatrix}-1\\1\end{bmatrix}$ (เพราะ $\begin{bmatrix}1(0)+2(1)-1(0)\\3(0)+4(1)-2(0)\end{bmatrix} = \begin{bmatrix}2\\4\end{bmatrix}$) $\to$ ใช้ Row operation :

$$\left[\begin{array}{cc|c}1&-1&2\\1&1&4\end{array}\right] \sim \left[\begin{array}{cc|c}1&-1&2\\0&2&2\end{array}\right]_{R_2-R_1} \sim \left[\begin{array}{cc|c}1&-1&2\\0&1&1\end{array}\right]_{\frac{1}{2}R_2} \sim \left[\begin{array}{cc|c}1&0&3\\0&1&1\end{array}\right]^{R_1-R_2}$$

ทำให้ได้ว่า $a_1 = 3$, $a_2 = 1$

ที่เรารู้ว่า $T\left(\begin{bmatrix}1\\0\\0\end{bmatrix}\right) = \begin{bmatrix}-1\\-2\end{bmatrix} = c_1\begin{bmatrix}1\\1\end{bmatrix} + c_2\begin{bmatrix}-1\\1\end{bmatrix}$ (เพราะ $\begin{bmatrix}1(1)+2(0)-1(0)\\3(1)+4(0)-2(0)\end{bmatrix} = \begin{bmatrix}1\\3\end{bmatrix}$) $\to$ ใช้ Row operation :

$$\left[\begin{array}{cc|c}1&-1&1\\1&1&3\end{array}\right] \sim \left[\begin{array}{cc|c}1&-1&1\\0&2&2\end{array}\right]_{R_2-R_1} \sim \left[\begin{array}{cc|c}1&-1&1\\0&1&1\end{array}\right]_{\frac{1}{2}R_2} \sim \left[\begin{array}{cc|c}1&0&2\\0&1&1\end{array}\right]^{R_1-R_2}$$

ทำให้ได้ว่า $a_1 = 2$, $a_2 = 1$

$\therefore$ $[T]_{B'B} = \begin{bmatrix}1/2 & 3 & 2\\ 1/2 & 1 & 1\end{bmatrix}$ # Ans.

4. Given a linear transformation $T : \mathbb{R}^4 \to \mathbb{R}^3$ defined by

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}\right) = \begin{bmatrix} 2x_2 - 5x_4 \\ x_1 + 3x_2 \\ x_3 \end{bmatrix}$$

Find bases for the kernel and range of this linear transformation. Also verify that the conclusion of the Rank-Nullity theorem holds for this T.

Sol จากโจทย์ หา $\text{Ker}(T) = \{\vec{x} \in \mathbb{R}^4 \mid T(\vec{x}) = \vec{0}_{\mathbb{R}^3}\}$ โดยให้ $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$; $\vec{x} \in \mathbb{R}^4$

จะได้ว่า $T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}\right) = \begin{bmatrix} 2x_2 - 5x_4 \\ x_1 + 3x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

ใช้ Row Operation : $\begin{bmatrix} 0 & 2 & 0 & -5 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 2 & 0 & -5 \\ 0 & 0 & 1 & 0 \end{bmatrix} \overset{R_{12}}{\sim} \begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 1 & 0 & -5/2 \\ 0 & 0 & 1 & 0 \end{bmatrix} \overset{\frac{1}{2}R_2}{\sim} \begin{bmatrix} 1 & 0 & 0 & 15/2 \\ 0 & 1 & 0 & -5/2 \\ 0 & 0 & 1 & 0 \end{bmatrix} \overset{R_1 - 3R_2}{}$

จะได้ว่า $x_1 + \frac{15}{2}x_4 = 0$, $x_2 - \frac{5}{2}x_4 = 0$, $x_3 = 0$ $\Rightarrow$ $x_1 = -\frac{15}{2}x_4$, $x_2 = \frac{5}{2}x_4$ โดยที่ $x_4 \in \mathbb{R}$ เป็น free variable

ทำให้ได้ว่า $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} (-15/2)x_4 \\ (5/2)x_4 \\ 0 \\ 1 \end{bmatrix} = x_4 \begin{bmatrix} -15/2 \\ 5/2 \\ 0 \\ 1 \end{bmatrix}$

นั่นคือ $\text{Ker}(T) = \left\{ x_4 \begin{bmatrix} -15/2 \\ 5/2 \\ 0 \\ 1 \end{bmatrix} \middle| \, x_4 \in \mathbb{R} \right\}$

$$\text{Ker}(T) = \text{Span}\left\{\begin{bmatrix} -15/2 \\ 5/2 \\ 0 \\ 1 \end{bmatrix}\right\} \#$$

ดังนั้น basis ของ $\text{Ker}(T)$ คือ $\left\{\begin{bmatrix} -15/2 \\ 5/2 \\ 0 \\ 1 \end{bmatrix}\right\}$ และจะได้ $\text{nullity}(T) = \dim(\text{Ker}(T)) = 1$ #

และจากนั้น หา $\text{range}(T) = \{\vec{x} \in \mathbb{R}^3 \mid \vec{x} = T(\vec{v}) \text{ for some } \vec{v} \in \mathbb{R}^3\}$

จะได้ว่า $\begin{bmatrix} 2x_2 - 5x_4 \\ x_1 + 3x_2 \\ x_3 \end{bmatrix} = x_1 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + x_4 \begin{bmatrix} -5 \\ 0 \\ 0 \end{bmatrix}$

ทำให้ได้ว่า $\text{range}(T) = \text{Span}\left\{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -5 \\ 0 \\ 0 \end{bmatrix}\right\}$

พิจารณา $\begin{bmatrix} -5 \\ 0 \\ 0 \end{bmatrix} = -\frac{15}{2}\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + \frac{5}{2}\begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} + 0\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

ส่งผลทำให้ $\text{range}(T) = \text{Span}\left\{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right\}$ # และจะได้ $\text{rank}(T) = \dim(\text{range}(T)) = 3$ #

หรืออีกกรณีคือ $\begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} = 3\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} - \frac{2}{5}\begin{bmatrix} -5 \\ 0 \\ 0 \end{bmatrix} + 0\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

ส่งผลทำให้ $\text{range}(T) = \text{Span}\left\{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -5 \\ 0 \\ 0 \end{bmatrix}\right\}$ # และจะได้ $\text{rank}(T) = \dim(\text{range}(T)) = 3$ #

ให้ได้ทั้งสองกรณี เพราะให้ค่า rank(T) เท่ากัน

$\therefore$ basis $\text{Ker}(T) = \left\{\begin{bmatrix} -15/2 \\ 5/2 \\ 0 \\ 1 \end{bmatrix}\right\}$, basis $\text{range}(T) = \left\{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right\}$ หรือ $\left\{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -5 \\ 0 \\ 0 \end{bmatrix}\right\}$,

$\text{nullity}(T) = 1$, และ $\text{rank}\, T = 3$ ## Ans.

5. Find the eigenvalues and associated eigenvectors of the given matrices.

a) $\begin{bmatrix} 0 & 3 \\ 3 & 0 \end{bmatrix}$

b) $\begin{bmatrix} 2 & 0 & -2 \\ 3 & 4 & 3 \\ 2 & 0 & 6 \end{bmatrix}$

Sol: a) จาก $(A-\lambda I_2)\vec{x} = \vec{0}$

จะได้ $\det(A-\lambda I_2) = 0$

$$\begin{vmatrix} -\lambda & 3 \\ 3 & -\lambda \end{vmatrix} = 0$$

$$\lambda^2 - 9 = 0$$

ดังนั้นได้ $\lambda = -3, 3$ # Ans.

กรณี $\lambda = -3$: $(A+3I)\vec{x} = \vec{0}$

$$\begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix}\vec{x} = \vec{0}$$

ดังนั้นได้ $\vec{x} = x_1\begin{bmatrix} 1 \\ -1 \end{bmatrix}$; $x_1 \in \mathbb{R}$

$\therefore$ eigenvector ของ $\lambda = -3$ คือ $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ # Ans.

กรณี $\lambda = 3$: $(A-3I)\vec{x} = \vec{0}$

$$\begin{bmatrix} -3 & 3 \\ 3 & -3 \end{bmatrix}\vec{x} = \vec{0}$$

ดังนั้นได้ $\vec{x} = x_1\begin{bmatrix} 1 \\ 1 \end{bmatrix}$; $x_1 \in \mathbb{R}$

$\therefore$ eigenvector ของ $\lambda = 3$ คือ $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ # Ans.

b) จาก $(A-\lambda I_3)\vec{x} = \vec{0}$

จะได้ $\det(A-\lambda I_3) = 0$

$$\begin{vmatrix} 2-\lambda & 0 & -2 \\ 3 & 4-\lambda & 3 \\ 2 & 0 & 6-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)(4-\lambda)(6-\lambda) + 4(4-\lambda) = 0$$

$$(4-\lambda)\left[(2-\lambda)(6-\lambda)+4\right] = 0$$

$$(4-\lambda)\left[12-8\lambda+\lambda^2+4\right] = 0$$

$$(4-\lambda)\left[\lambda^2-8\lambda+16\right] = 0$$

$$(4-\lambda)(\lambda-4)^2 = 0$$

ดังนั้นได้ $\lambda = 4$ ที่มีการซ้ำกันทั้งหมด 3 ครั้ง

กรณี $\lambda = 4$: $(A-4I)\vec{x} = \vec{0}$

$$\begin{bmatrix} -2 & 0 & -2 \\ 3 & 0 & 3 \\ 2 & 0 & 2 \end{bmatrix}\vec{x} = \vec{0}$$

ดังนั้นได้ $\vec{x} = x_1\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + x_2\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$; $x_1, x_2 \in \mathbb{R}$

$\therefore$ eigenvector ของ $\lambda = 4$ คือ $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ # Ans.

6. Show that the eigenvalues of a diagonal matrix are the numbers on the diagonal.

Sol: ให้ D เป็นเมทริกซ์แนวทแยงมุมอยู่ในรูป $D = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix}$ โดย $d_1, d_2, \ldots, d_n$ เป็นสมาชิกในแนวทแยงมุม

จากค่า eigenvalue ของเมทริกซ์ A หาได้จาก charateristic equation : $\det(A-\lambda I)=0$ โดย λ = eigenvalue, I = เมทริกซ์เอกลักษณ์

ทำให้ได้ว่า

$$\det(D-\lambda I) = \begin{vmatrix} d_1-\lambda & 0 & \cdots & 0 \\ 0 & d_2-\lambda & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & d_n-\lambda \end{vmatrix} = (d_1-\lambda)(d_2-\lambda)\ldots(d_n-\lambda)$$

จะได้ charateristic polynomial $(d_1-\lambda)(d_2-\lambda)\ldots(d_n-\lambda) = 0$

จากสมการข้างต้นจะเป็นจริง ก็ต่อเมื่อ $\lambda = d_1, d_2, \ldots, d_n$ เท่านั้น

ดังนั้น eigenvalue ของเมทริกซ์แนวทแยงมุม มีจำนวนเท่ากับ จำนวนสมาชิกในแนวทแยงมุม // Ans.

7. Show that if λ is an eigenvalue of A and k is a positive integer, then λ^k is an eigenvalue of A^k.

Sol: จะได้ว่า $Ax = \lambda x$

$\times A$; $A^2x = A\lambda x$

$A^2x = \lambda(Ax)$ แทน Ax ด้วย λx

$A^2x = \lambda^2 x$

$\times A$; $A^3x = A\lambda^2 x$

$A^3x = \lambda^2(Ax)$ แทน Ax ด้วย λx

$A^3x = \lambda^3 x$

$\therefore$ ทำให้ได้ว่า $A^kx = \lambda^k x$ // Ans.

8. Compute the eigenspace for each eigenvalue of $A = \begin{bmatrix} 5 & -1 & 1 \\ 0 & 4 & 0 \\ 1 & -1 & 5 \end{bmatrix}$.

Also give the multiplicities of each eigenvalue.

Sol: จะได้ว่า $\det(A-\lambda I) = \begin{vmatrix} 5-\lambda & -1 & 1 \\ 0 & 4-\lambda & 0 \\ 1 & -1 & 5-\lambda \end{vmatrix}$

$(5-\lambda)^2(4-\lambda)-(4-\lambda) = 0$

$(4-\lambda)[(5-\lambda)^2-1] = 0$

$(4-\lambda)[\lambda^2-10\lambda+25-1] = 0$

$(4-\lambda)[\lambda^2-10\lambda+24] = 0$

$(4-\lambda)(\lambda-6)(\lambda-4) = 0$

จะได้ $\lambda = 4$ ที่มีภาวะซ้ำกัน 2 ครั้ง

$\lambda = 6$ ที่มีภาวะซ้ำกัน 1 ครั้ง

กรณี $\lambda=4$: $(A-4I)\vec{x} = \vec{0}$

$\begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 0 \\ 1 & -1 & 1 \end{bmatrix}$

9. Determine whether the matrix A is diagonalizable. If it is, find an invertible matrix P such that $P^{-1}AP$ is a diagonal matrix and determine $P^{-1}AP$.

a) $\begin{bmatrix} 4 & 1 \\ -1 & 2 \end{bmatrix}$

b) $\begin{bmatrix} 2 & 0 & 1 \\ 0 & 5 & 0 \\ 3 & 0 & 4 \end{bmatrix}$

Soln

10. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear transformation defined by

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 3 & 4 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Find a basis for $\mathbb{R}^2$ for which the matrix representation for T is diagonal.

Soln:

Keys to Exercises

Exercise 1

1. $x = -4, y = 3, w = 3, z = 0$.

2.

$$\left[\begin{array}{ccc|c} 2 & -1 & 0 & 0 \\ -1 & 2 & 1 & 6 \\ -1 & 0 & -1 & -4 \end{array}\right] \xrightarrow{R_{13}} \left[\begin{array}{ccc|c} -1 & 0 & -1 & -4 \\ -1 & 2 & 1 & 6 \\ 2 & -1 & 0 & 0 \end{array}\right] \xrightarrow[R_1+R_2,-2R_1+R_3]{-R_1} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 4 \\ 0 & 2 & 2 & 10 \\ 0 & -1 & -2 & -8 \end{array}\right]$$
$$\xrightarrow{\frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 4 \\ 0 & 1 & 1 & 5 \\ 0 & -1 & -2 & -8 \end{array}\right] \xrightarrow{R_3+R_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 4 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & -1 & -3 \end{array}\right]$$

Thus, $x_3 = 3, x_2 = 2, x_1 = 1$.

3.

$$\left[\begin{array}{ccc|c} 2 & 6 & 1 & 5 \\ 3 & 9 & 2 & 8 \\ 0 & -1 & 3 & 3 \end{array}\right] \xrightarrow{-R_1+R_2} \left[\begin{array}{ccc|c} 2 & 6 & 1 & 5 \\ 1 & 3 & 1 & 3 \\ 0 & -1 & 3 & 3 \end{array}\right] \xrightarrow{R_{12}} \left[\begin{array}{ccc|c} 1 & 3 & 1 & 3 \\ 2 & 6 & 1 & 5 \\ 0 & -1 & 3 & 3 \end{array}\right]$$
$$\xrightarrow{-2R_1+R_2} \left[\begin{array}{ccc|c} 1 & 3 & 1 & 3 \\ 0 & 0 & -1 & -1 \\ 0 & -1 & 3 & 3 \end{array}\right] \xrightarrow{R_{23}} \left[\begin{array}{ccc|c} 1 & 3 & 1 & 3 \\ 0 & -1 & 3 & 3 \\ 0 & 0 & -1 & -1 \end{array}\right]$$
$$\xrightarrow[-R_3]{-R_2} \left[\begin{array}{ccc|c} 1 & 3 & 1 & 3 \\ 0 & 1 & -3 & -3 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$x_3 = 1, x_2 = 0$ and $x_1 = 2$.

4.

$$\left[\begin{array}{ccc|c} 0 & 2 & 5 & 15 \\ -1 & 7 & -5 & -20 \\ -1 & 8 & 3 & 4 \end{array}\right] \xrightarrow{R_{12}} \left[\begin{array}{ccc|c} -1 & 7 & -5 & -20 \\ 0 & 2 & 5 & 15 \\ -1 & 8 & 3 & 4 \end{array}\right] \xrightarrow{-R_1+R_3} \left[\begin{array}{ccc|c} -1 & 7 & -5 & -20 \\ 0 & 2 & 5 & 15 \\ 0 & 1 & 8 & 24 \end{array}\right]$$

$$\xrightarrow{R_{23}} \left[\begin{array}{ccc|c} -1 & 7 & -5 & -20 \\ 0 & 1 & 8 & 24 \\ 0 & 2 & 5 & 15 \end{array}\right] \xrightarrow{-2R_2+R_3} \left[\begin{array}{ccc|c} -1 & 7 & -5 & -20 \\ 0 & 1 & 8 & 24 \\ 0 & 0 & -11 & -33 \end{array}\right]$$

$x_3 = 3, x_2 = 0$ and $x_1 = 5$.

5. $-2/3R_1 + R_2, -1/3R_1 + R_3, -5/3R_2 + R_3$

6. Consistent: $x_1 = 1 - 2x_3, x_2 = -1 + 3x_3$ and $x_3 = x_3$ a free variable.

7.

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 3 \\ 1 & -1 & 3 & 4 \\ 1 & 1 & a^2 - 10 & a \end{array}\right] \xrightarrow[-R_1+R_3]{-R_1+R_2} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 3 \\ 0 & -2 & 4 & 1 \\ 0 & 0 & a^2 - 9 & a - 3 \end{array}\right]$$

The system has no solution when $a = -3$, it has the unique solution when $a \neq \pm 3$ and has infinitely many solutions when $a = 3$.

Exercise 2

1. Clear $\begin{bmatrix} 0 \\ 0 \end{bmatrix} \in U$, so $U \neq \phi$. Let $\begin{bmatrix} x_1 \\ y_1 \end{bmatrix}, \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} \in U$ and $c \in \mathbb{R}$. Then $2(cx_1 + x_2) + 3(cy_1 + y_2) = c(2x_1 + 3y_1) + (2x_2 + 3y_2) = 0$. Thus, $c \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} \in U$.

2. a) linearly independent. b) linearly independent.

3.

$$\begin{bmatrix} 0 & a & 1 \\ 1 & 1 & a \\ a & 0 & 1 \end{bmatrix} \xrightarrow{R_{12}} \begin{bmatrix} 1 & 1 & a \\ 0 & a & 1 \\ a & 0 & 1 \end{bmatrix} \xrightarrow{-aR_1+R_3} \begin{bmatrix} 1 & 1 & a \\ 0 & a & 1 \\ 0 & -a & 1 - a^2 \end{bmatrix}$$

$$\xrightarrow{R_2+R_3} \begin{bmatrix} 1 & 1 & a \\ 0 & a & 1 \\ 0 & 0 & 2 - a^2 \end{bmatrix}$$

$a \neq 0, \pm\sqrt{2}$.

4. $v = 2\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$, so $v \in Span(X)$, but u. Moreover, $\alpha = -2$.

5. It is easy to see that the set is linearly independent. The vector $u = x_1 + x_2 + \cdots + x_{n-1}$.

6. A basis is $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and the dimension is 1.

7. A basis is $\left\{ \begin{bmatrix} 7 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -10 \\ 0 \\ 1 \end{bmatrix} \right\}$ and the dimension is 2.

8. It is easy to see that the set is linearly independent. An extended basis is $\left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \right\}$.

9. $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$.

10. rank = 2 and nullity = 1.

Exercise 3

1. a) Yes. It can be seen that $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2 & 1 \\ 1 & -3 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$.

   b) No. In general, $\cos(x + y) \neq \cos x + \cos y$.

2. a) $T\left(\begin{bmatrix} -2 \\ 4 \end{bmatrix}\right) = T\left(-2\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 4\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = T\left(-2\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) + T\left(4\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = -2\begin{bmatrix} 1 \\ 3 \end{bmatrix} + 4\begin{bmatrix} -2 \\ 4 \end{bmatrix} = \begin{bmatrix} -10 \\ 10 \end{bmatrix}$.

b) $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = T\left(x\begin{bmatrix} 1 \\ 0 \end{bmatrix} + y\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = T\left(x\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) + T\left(y\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = x\begin{bmatrix} 1 \\ 3 \end{bmatrix} + y\begin{bmatrix} -2 \\ 4 \end{bmatrix} = \begin{bmatrix} x - 2y \\ 3x + 4y \end{bmatrix}$.

c) $A = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$.

3. $\begin{bmatrix} 1/2 & 3 & 2 \\ 1/2 & 1 & 1 \end{bmatrix}$.

4. A basis for the kernel is $\left\{\begin{bmatrix} -7.5 \\ 2.5 \\ 0 \\ 1 \end{bmatrix}\right\}$ and a basis for the range is $\left\{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right\}$.
This is clear that it satisfies the Rank-Nullity Theorem.

5. a) $v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ with eigenvalue $\lambda_1 = 3$ and $v = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ with eigenvalue $\lambda_2 = -3$.

b)

c) $v = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ with eigenvalue $\lambda_1 = 4$ and $v = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ with eigenvalue $\lambda_1 = 4$.

6. Let $D = Diag(d_1, d_2, \ldots, d_n)$. Then characteristic polynomial is $(x - d_1)(x - d_2)\cdots(x - d_n) = 0$. The eigenvalues of D are $d_1, d_2, \cdots, d_n$.

7. Let v be an eigenvector with corresponding eigenvalue λ. That is, $Av = \lambda v$. Then $A^k v = A^{k-1}(Av) = A^{k-1}\lambda v = \lambda A^{k-2}(Av) = \lambda^2 A^{k-2} v = \cdots = \lambda^k v$.

8. Eigenspace $span\left\{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}\right\}$ with corresponding eigenvalue 6 and the multiplicity 1.

Eigenspace $span\left\{\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}\right\}$ with corresponding eigenvalue 4 and the multiplicity 2.

9. a) The matrix is not diagonalizable since it admits only one eigenvector.

   b) The matrix is diagonalizable with $P = \begin{bmatrix} -1 & 0 & 1/3 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ and $D = Diag(1, 5, 5)$.

10. The matrix for the map T is diagonalizable with $P = \begin{bmatrix} 2 & -2/3 \\ 1 & 1 \end{bmatrix}$. Therefore we can choose a basis $\left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -2/3 \\ 1 \end{bmatrix} \right\}$.